BMS MONITOR

Engineering Review Pack

Prototype SDLC evidence, SRS, and production-readiness gates

Review status	READY FOR CONTROLLED PROTOTYPE REVIEW
Decision requested	Approve demo/static review only; do not connect a real battery yet.
Implementation baseline	Flutter Web / Dart / Provider / Flutter 3.47.0
Evidence baseline	Supplied `bmsdash.zip`, repository source, tests, scripts, and project documentation
Prepared for	Company product, engineering, QA, security, battery-domain, and release review

Prepared from the supplied project package; unsupported approvals and deployments remain explicitly open.

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How to use this pack

This is an implementation-grounded review document. It separates what is present in the ZIP from what requires company action. A green status means evidence exists for the prototype scope; amber means a decision, owner, or external dependency is still required; red means the item must block real-gateway use.

1 1. Review decision

Recommended decision

Approve the supplied package for controlled prototype/demo review and conditional static deployment.

Do **not** approve it as a production battery-monitoring, safety, or battery-control system. Do **not** connect it to a real battery until the gateway contract, server-side security boundary, chemistry-specific thresholds, operating ownership, and staging acceptance gates are closed.

1.1 What the company is reviewing

The deliverable is a responsive, read-only Flutter Web dashboard for presenting battery-management telemetry on industrial landscape displays, tablets, and mobile browsers. It provides Overview, Faults, Trends, Cells, and Settings views. The current default is deliberately simulated data; the application does not claim that a physical battery is connected.

Decision area	Evidence in supplied package	Review conclusion
Product / UX	Responsive shell, telemetry cards, navigation, industrial-console visual direction	PROTOTYPE READY
Engineering	Typed telemetry model, provider lifecycle, defensive parser, WebSocket seam	PROTOTYPE READY
Quality	Nine test cases in `test/`, analyzer/build commands, release checklist	EVIDENCED
Security	No secrets in package; `wss://` rule and headers; no auth or backend	GATE REQUIRED
Battery safety	Read-only boundary; prototype alert thresholds only	BLOCKS LIVE USE
Company release	Checklist exists; no repository URL, PR, merge SHA, tag, or deployment proof in ZIP	COMPANY ACTION

1.2 Executive summary

The package is strong as a reviewable UI and software-boundary prototype. It demonstrates the information architecture, visual hierarchy, demo/live/stale semantics, typed parsing, bounded payload handling, reconnect behavior, and a credible path to a secure staging architecture. Its most important product decision is also its safest one: control commands are absent by design.

The package is not evidence of a production service. Missing or unapproved items include the actual gateway protocol and schema owner, authentication and authorization, time-series persistence, alert escalation, battery-specific operating limits, operational ownership, device/browser acceptance, penetration testing, disaster recovery, and company Git/release records.

One documentation inconsistency to resolve

`plan.md` says "all five unit/widget tests pass," while the repository contains nine test declarations and `README.md` says nine tests. The implementation evidence supports nine test cases; update `plan.md` or attach the exact CI transcript before sign-off.

2 2. Product boundary and intended use

2.1 Product definition

A browser dashboard that presents simulated or validated **read-only** BMS telemetry with explicit demo, live, stale, and unavailable states. It is a monitoring surface, not a BMS controller, protection relay, safety instrument, or substitute for manufacturer controls.

2.2 In scope for the current prototype

- Overview of SOC, SOH, pack voltage, current, power, energy, capacity, cycle count, operating mode, and temperatures.
- Faults and recent events, including prototype low-SOC, low-SOH, high-temperature, and stale-data alerts.
- Trends with illustrative 1H / 6H / 24H sample histories; these are not a historical backend.
- Cells and balancing presentation where the payload supplies cell information.
- Settings and simulated-warning interaction for review.
- Demo telemetry by default and an opt-in browser WebSocket boundary for validated snapshots.
- Static Flutter Web output configured for Vercel, with route metadata and security headers.

2.3 Explicitly out of scope

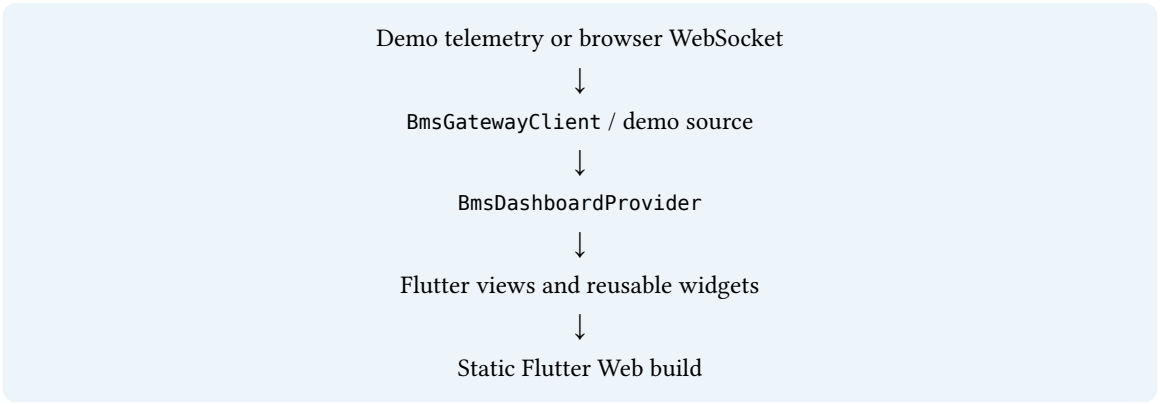
Battery charging/discharging, contactor control, reset, configuration writes, remote commands, private gateway credentials in browser code, certified operating limits, user roles, historical storage, external notifications, MQTT/CAN/Modbus adapters, regulatory certification, and physical battery safety validation.

Non-negotiable safety boundary

Any value passed through Flutter Web build variables is inspectable by users. Private tokens, passwords, certificates, and gateway secrets must not be placed in `--dart-define`, Vercel frontend variables, or committed source. Real integration must use an authenticated server-side gateway/proxy.

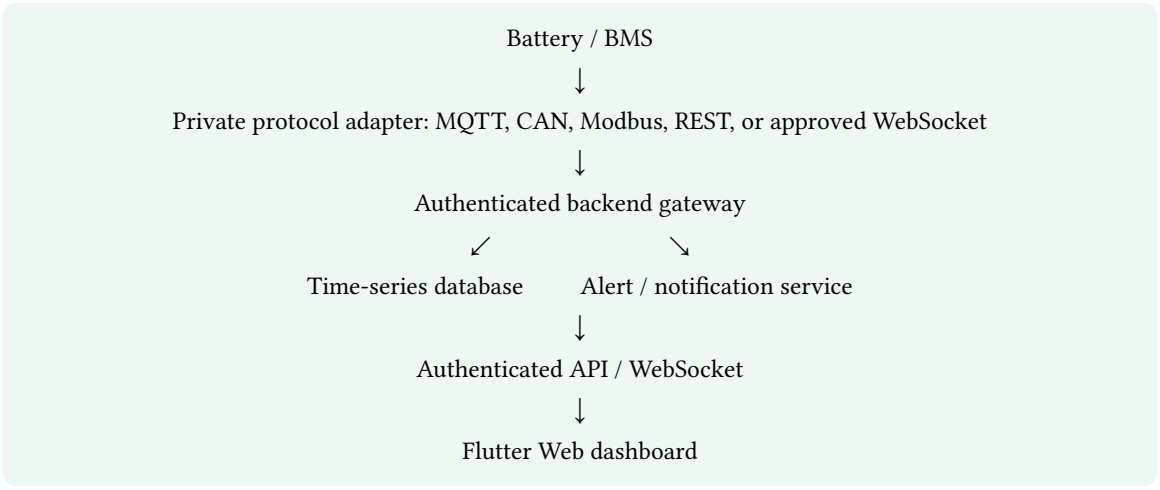
3 3. Current-state architecture

3.1 Implemented prototype flow



Component	Responsibility demonstrated by the ZIP	Source evidence
`BmsTelemetry`	Typed measurements, cells, events, parsing, bounds, derived alerts	`lib/models/bms_telemetry.dart`
Demo source	Deterministic sample readings and clearly labeled demo identity	`lib/data/demo_telemetry.dart`
Gateway client	URL validation, stream handling, message-size bound, JSON parsing	`lib/services/gateway_client.dart`
Dashboard provider	Lifecycle, bounded reconnects, stale state, last-reading preservation, source label	`lib/services/bms_dashboard_provider.dart`
Views/widgets	Render state only; no battery commands	`lib/views/`, `lib/widgets/`
Release path	Pinned SDK bootstrap, Vercel build/output configuration, route manifest	`scripts/`, `vercel.json`, `web/`

3.2 Target production architecture



The browser must not connect directly to a private battery network. The backend must own device registration, authentication, authorization, schema versioning, rate limiting, audit logs, threshold configuration, retention, escalation, monitoring, and recovery.

4 4. Software requirements specification

The following requirements are intentionally written as review decisions rather than generic feature statements. “Implemented” means implemented for the prototype scope; it is not a production certification claim.

4.1 Functional requirements

ID	Requirement / acceptance intent	Current result	Acceptance owner
FR-001	Overview presents current pack condition with clear hierarchy	Complete	Product owner
FR-002	SOC is shown as percentage with segmented visual treatment and units	Complete	UX reviewer
FR-003	Pack voltage, current, power, energy, capacity, SOH, temperatures, and mode are shown with units	Complete	Product owner
FR-004	Unknown/fault operating modes are not styled as healthy	Complete	QA lead
FR-005	Faults, warnings, and recent events are visible and ordered for review	Complete	Operations
FR-006	Overview, Faults, Trends, Cells, and Settings are reachable through persistent navigation	Complete	Product owner
FR-007	Simulated values are visibly labeled when no approved live source exists	Complete	Security reviewer
FR-008	Validated read-only WebSocket snapshots can be accepted when configured	Partial	Platform engineer
FR-009	Malformed, incomplete, implausible, or oversized snapshots are rejected	Complete	QA lead
FR-010	Disconnect preserves the last reading and marks it stale; reconnect is bounded	Complete	QA lead
FR-011	Historical telemetry is stored and queryable for production use	Not built	Platform engineer
FR-012	Authenticated viewer/operator/admin roles are enforced	Not built	Security reviewer
FR-013	External notifications follow an approved escalation policy	Not built	Operations owner
FR-014	Battery-control commands are exposed	Out of scope	Safety authority

4.2 Non-functional requirements and current evidence

ID	Requirement	Evidence / gap
NFR-001	Responsive layout across industrial landscape, tablet, and mobile widths	Layout builders exist; formal device matrix still pending
NFR-002	High-contrast industrial presentation with clear semantic status colors	Theme tokens and component styling are present
NFR-003	Never present demo values as verified live values	Demo/source labels implemented

NFR-004	Bound untrusted input and collection sizes	256 KB message; 500 cells; 200 events; 100 alerts; 100 zones
NFR-005	Require `wss://` from HTTPS browser sessions	Implemented in gateway adapter
NFR-006	Reproducible build toolchain	Flutter 3.47.0 pinned; SHA-256 bootstrap in script
NFR-007	Static analysis, tests, and release build remain clean	Repository claims pass; CI/company transcript not supplied
NFR-008	Production telemetry is confidential and authenticated	Not met by static frontend; backend required
NFR-009	Availability, backup, recovery, monitoring, and incident path	Not implemented; operations design required
NFR-010	Thresholds reflect approved chemistry/manufacturer limits	Blocked until battery-domain input is supplied

5 5. Data contract and integration gate

5.1 Validated snapshot boundary

Required fields are `soc`, `packVoltageV`, `packCurrentA`, `powerKw`, `energyTodayKwh`, `remainingCapacityKwh`, `sohPercent`, `nominalCapacityKwh`, `cycleCount`, `operatingMode`, `minTempC`, `avgTempC`, `maxTempC`, and `lastBalanced`. Optional fields include gateway identity, timestamp, temperature zones, cells, events, alerts, warnings, and critical faults.

The parser rejects missing required fields, invalid SOC/SOH ranges, implausible electrical values, invalid cycle counts, impossible temperature ordering, unknown operating modes, oversized messages, and oversized collections. These are defensive software constraints, not approved operating limits for any chemistry or pack.

5.2 Inputs required before live staging

Input	What must be supplied	Decision owner
Gateway identity	Manufacturer/model, gateway model, device identity, environment	Battery / platform
Protocol	Transport, schema, versioning, sample payload, error/heartbeat behavior	Platform engineer
Security	Auth method, TLS requirements, allowed origins, secret storage, session model	Security reviewer
Battery context	Chemistry, manufacturer, pack limits, approved alarm thresholds	Battery engineer
Operations	Alert routing, on-call owner, stale-data response, retention, incident path	Operations owner
Test access	Non-production endpoint, synthetic data, device/browser matrix	QA / release

Integration rule

The current browser WebSocket seam is suitable for a browser-safe read-only staging gateway only after security review. It is not a production credential boundary and must not be pointed directly at a private battery network.

6 6. Verification evidence from the supplied ZIP

6.1 Checks represented by the repository

Check	Recorded result	Review interpretation
`dart format lib test`	Passed	Formatting evidence is described in the package
`flutter test`	9 test cases	Nine declarations are present across three test files; attach CI output for final audit evidence
`flutter analyze`	No issues	Static analysis claim in README/plan; rerun in company CI
`flutter build web --release`	Passed	Release build claim in project docs; output not included in ZIP
`vercel.json` parse	Passed	Configuration is valid JSON and names the project build/output path
Route manifest	Passed locally	`/manus-routes.json` is included under `web/`
ZIP integrity / cleanliness	Passed	Package excludes `.dart_tool/`, build output, `.git`, and logs

6.2 Test intent represented in source

- Telemetry conversion, event ordering, required-field/range rejection, and bounded collection handling.
- Derived low-health and high-temperature alerts.
- Navigation between dashboard views and simulated-warning behavior.
- Invalid gateway configuration fallback to clearly labeled demo data.
- Provider lifecycle and stale-data behavior.

6.3 Not yet verified by the supplied package

Real gateway contract tests; browser/device acceptance on company hardware; backend integration; authentication and authorization; penetration/security testing; load testing; notification delivery; disaster recovery; accessibility sign-off; and physical battery safety validation.

Evidence discipline

A ZIP file is a review package, not a Git commit or deployment record. The company review should record the repository URL, accepted commit SHA, PR/approval trail where applicable, CI run URL, release tag, deployed URL, rollback owner, and artifact checksum.

7 7. SDLC execution status

The project follows Agile incremental delivery with requirement-linked verification and DevOps release practices. The handwritten lifecycle intent is normalized below into company review gates.

No.	Lifecycle stage	Project-specific evidence / exit condition	Status
1	Requirements understanding	Scope, screens, metrics, safety boundary, and missing gateway inputs are documented	Met
2	Cross-functional planning	Product, operations, battery, frontend, platform, security, QA, and release concerns identified	Met
3	Architecture understanding	Prototype flow and recommended backend boundary documented; no command path in UI	Met / next gate
4	Design	Industrial telemetry-console visual system, responsive layout, navigation and semantic states implemented	Prototype met
5	Implementation	Typed model, demo source, provider, views, widgets, build scripts and docs included	Met
6	Testing	Nine test declarations and documented analyzer/ build checks; company CI evidence still required	Partially evidenced
7	Version control	Lockfile, pinned SDK, scripts, clean ZIP; company commit/tag/branch controls not supplied	Company action
8	PR / merge	Checklist exists; no PR URL, approvals, CI transcript, merge SHA, or branch protection record	Not evidenced
9	Documentation / handoff	README, architecture, gateway, security, limitations, deployment, contributing, release docs	Met for prototype

8 8. Risk register and ownership

ID	Risk	Impact	Current control / required action
R-001	Frontend exposes gateway credentials	Unauthorized access or device compromise	No credentials in source; require backend proxy and secret storage before live use
R-002	Prototype thresholds are wrong for actual battery	Missed/noisy alarms or unsafe decisions	Labels identify prototype; battery engineer must approve chemistry-specific limits
R-003	Gateway schema changes	Parse failures or misleading values	Strict parser and bounds; add versioned contract tests
R-004	Disconnect hides old data	Operator acts on stale state	Stale marker and last-reading preservation; add heartbeat/ monitoring in backend

R-005	Illustrative trends mistaken for history	Incorrect operational interpretation	Label as illustrative; replace with time-series data before operational use
R-006	No identity, audit trail, or escalation	Uncontrolled access and missed incidents	Prototype is static/read-only; add sessions, roles, audit, alerts, on-call workflow
R-007	Company release evidence missing	Cannot prove accepted version or rollback point	Record repository, commit SHA, PR/approvals, CI, tag, URL, owner and checksum
R-008	Responsive/accessibility behavior not formally accepted	Poor use on target devices	Run device matrix, keyboard/ focus, contrast and reduced-motion review

9 9. Path to real-gateway staging

The safest next increment is a secure read-only staging platform. It should preserve the current dashboard and replace the direct browser seam with a controlled backend boundary.

Phase	Outcome	Exit criteria	Gate owner
P0	Prototype sign-off	Product owner accepts scope; demo label, read-only boundary, and documentation inconsistency resolved	Product
P1	Company source/release record	Repository, accepted commit, CI run, review approvals, tag, checksum, and deployment record captured	Release / QA
P2	Gateway contract baseline	Protocol/schema/version, sample payloads, heartbeat, errors, device identity, and non-prod endpoint approved	Platform / battery
P3	Security and operations design	Backend proxy, auth/roles, TLS, secrets, rate limits, audit, alert routing, retention, monitoring, rollback and incident path approved	Security / operations
P4	Read-only staging	Synthetic or non-production gateway connected; contract, stale, reconnect, access, device, and notification tests pass	QA / battery
P5	Operational readiness review	Thresholds approved, runbook signed, on-call assigned, accessibility/device acceptance complete	Management / safety

Phase constraint

Do not add charging, discharging, contactor, reset, or remote-configuration commands in the next increment. Any future control surface requires a separate safety and authorization architecture, threat model, domain review, and approval process.

10 10. Company review checklist

No.	Review action	Status
1	Confirm product owner accepts “read-only prototype” scope and demo/static deployment decision	Open
2	Resolve `plan.md` five-test wording vs nine test declarations and attach CI transcript	Open
3	Confirm repository owner, branch, accepted commit SHA, PR/approval record, merge/tag, and checksum	Open
4	Complete manual UI review: all five views, demo labels, fault/stale states, tablet/mobile widths, no commands	Open
5	Assign battery-domain owner and approve actual chemistry/manufacturer thresholds	Blocked for live
6	Approve gateway contract, non-production endpoint, authentication, TLS, and device identity	Blocked for live
7	Approve backend, identity, audit, monitoring, alert escalation, backup/recovery, and incident model	Blocked for live
8	Record deployment URL, release owner, logs, rollback path, and operational handoff	Open

11 11. Artifact register

Artifact	Review use
`README.md`	Setup, status, demo limitation, gateway boundary, validation and deployment notes
`plan.md`	Lifecycle model, implementation approach, delivery constraints; update test-count wording
`docs/ARCHITECTURE.md`	Implemented prototype flow and recommended production architecture
`docs/GATEWAY-INTEGRATION.md`	Required fields, example snapshot, defensive limits, live-integration prerequisites
`docs/SECURITY.md`	Current protections, public-client limitation, required production controls
`docs/KNOWN-LIMITATIONS.md`	Explicit gaps and follow-on ownership
`docs/RELEASE-CHECKLIST.md`	Source, automated, UI, and deployment review actions
`lib/models/bms_telemetry.dart`	Typed telemetry contract and defensive parsing
`lib/services/bms_dashboard_provider.dart`	Lifecycle, reconnect, stale state and source labeling
`test/`	Nine test declarations across telemetry, navigation and provider behavior
`scripts/vercel_build.sh`, `scripts/bootstrap_flutter.sh`	Reproducible SDK/build path

`vercel.json`, `web/manus-routes.json`	Static deployment configuration and route manifest

12 12. Final conclusion

The BMS Monitor Dashboard is a credible, implementation-backed prototype for company review. Its strongest qualities are the explicit demo-data default, the absence of unsafe control commands, the typed telemetry boundary, the defensive parser, the stale-data semantics, and the documentation that distinguishes current behavior from production requirements. The correct company conclusion is conditional: accept the current package as a prototype and review artifact; do not represent it as production-ready or safety-certified. The project becomes eligible for real-gateway staging only after the company completes the open ownership, security, gateway, threshold, operational, device-acceptance, and release-evidence gates in this report.

Sign-off record

Product owner: _____

Date: _____

Engineering owner: _____

Date: _____

Battery-domain reviewer: _____

Date: _____

Security reviewer: _____

Date: _____

QA / release owner: _____

Date: _____

Management decision: ☐ Prototype approved ☐ More work required ☐ Live integration approved

End of engineering review pack — generated from the supplied BMS Monitor Dashboard project package.